Decent Coaching Classes

VIII SB

Mind Your Math - IX

Time: 2 Hrs **Marks: 60**

Note: (1) All questions are compulsory. (2) Figures to the right indicate full marks. (3) Draw neat and proper figures wherever necessary.

Q.1 (A) Choose the correct alternative for each of the following questions. **[8]**

(1) The orthocentre of an obtuse-angled triangle lies ___

(A) inside the triangle (B) on the longest side
(C) outside the triangle (D) at the vertex of the obtuse angle

(2) G is the centroid of ΔABC and seg BE is a median. If l(GE) = 3 cm, then l(BE) = ___

(A) 6 cm (B) 9 cm (C) 12 cm (D) 4.5 cm

(3) Which of the following points always lies in the interior of every triangle?

(A) Orthocentre (B) Circumcentre (C) Centroid (D) None of these

(4) The factors of $x^2 - 7x - 18$ are ___

(A) $(x - 9)(x + 2)$ (B) $(x + 9)(x - 2)$ (C) $(x - 6)(x - 3)$ (D) $(x - 18)(x + 1)$

(5) One factor of $8x^3 + 27$ is $(2x + 3)$. The other factor is ___

(A) $4x^2 - 6x + 9$ (B) $4x^2 + 6x + 9$ (C) $2x^2 - 6x + 9$ (D) $4x^2 - 12x + 9$

(6) $(x^2 - 4) / (x^2 - x - 2)$ in the simplest form is ___

(A) $(x + 2)/(x + 1)$ (B) $(x - 2)/(x + 1)$ (C) $(x + 2)/(x - 1)$ (D) $(x + 1)/(x + 2)$

(7) x varies directly as y. If x = 20 when y = 5, the constant of variation is ___

(A) 4 (B) 1/4 (C) 100 (D) 25

(8) Which of the following pairs is in inverse variation?

(A) Number of pens and their cost (B) Speed and time for a fixed distance
(C) Radius of a circle and its circumference (D) Weight of sugar and its price

Q.1 (B) Fill in the blanks (1, 2, 3, 5, 7) and state whether the following are true or false (4, 6, 8). **[8]**

(1) The segment joining a vertex of a triangle and the midpoint of the opposite side is called a _____ of the triangle.

(2) The perpendicular segment drawn from a vertex of a triangle to the line containing the opposite side is called an _____.

(3) G is the centroid of ΔXYZ and seg XP is a median. If l(XP) = 15 cm, then l(GP) = _____ cm.

(4) The three altitudes of a triangle are concurrent. (True / False)

(5) $a^3 + b^3 = (a + b)$(_____).

(6) $(x + 4)$ is a factor of $x^2 - x - 12$. (True / False)

(7) If x varies inversely as y, then $x \times y$ is a _____.

(8) The number of workers and the time taken to complete a job are in direct variation. (True / False)

Q.2 Solve any FIVE of the following sub-questions. **[10]**

(1) G is the centroid of ΔPQR and seg PM is a median. If l(PG) = 8 cm, find l(GM) and l(PM).

(2) In ΔABC, seg AD and seg BE are medians and G is their point of concurrence. If l(GE) = 2.5 cm, find l(BE).

(3) Factorise: $x^2 - 10x + 21$

(4) Factorise: $3y^2 + 14y + 8$

(5) Factorise: $27p^3 + 64$

(6) a varies directly as b. When a = 18, b = 6. Find a when b = 15.

(7) x varies inversely as y. When x = 24, y = 5. Find x when y = 8.

Q.3 Solve any FOUR of the following sub-questions. [12]

(1) Draw an acute-angled ΔPQR. Draw all its altitudes and name their point of concurrence as O.

(2) Draw an obtuse-angled ΔSTV. Draw all its medians and show their point of concurrence as G.

(3) Factorise: $2x^2 - 5x - 12$

(4) Factorise: $250p^3 - 128$

(5) y varies directly as the square of x. When x = 5, y = 50. Find y when x = 8.

(6) A tank is filled by 4 identical pipes in 9 hours. How much time will 6 such pipes take to fill the same tank?

Q.4 Solve any THREE of the following sub-questions. [12]

(1) In ΔABC, seg AD, seg BE and seg CF are medians and G is the centroid. If l(AG) = 10 cm, l(GE) = 4 cm and l(CF) = 21 cm, find (i) l(GD) (ii) l(BG) (iii) l(BE) (iv) l(CG).

(2) Simplify: $(m^2 - 5m + 6)/(m^2 - m - 2) \times (m^2 + m)/(m^2 - 9)$

(3) Simplify: $(3p + 2q)^3 - (3p - 2q)^3$

(4) p varies inversely as the square of q. When q = 3, p = 8. Find (i) p when q = 6, (ii) q when p = 18.

(5) A bus travelling at 40 km/hr takes 9 hours to complete a journey. (i) How much time will it take at 60 km/hr? (ii) At what speed should it travel to complete the journey in 7.5 hours?

Q.5 Solve any TWO of the following sub-questions. [10]

(1) Draw an equilateral triangle PQR of side 5 cm. Draw all its medians and altitudes. Write your observation about the centroid and the orthocentre of the triangle.

(2) (i) Factorise: $12x^2 - x - 35$. (ii) Factorise: $16a^3 - 250$.

(3) 18 workers can finish building a road in 20 days. (i) Find the constant of variation. (ii) How many workers are needed to finish the road in 12 days? (iii) In how many days will 24 workers finish the road?

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